

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Status by step — what's done, what broke, and how we fixed it

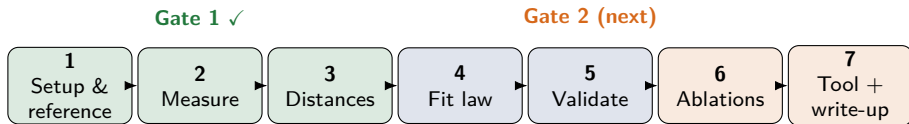
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Progress against the seven-step plan



DONE

done & validated

BUILT

code built, awaiting the run

NEXT

not

started

The measurement half (steps 1–3) is finished and **Gate 1 is passed**; the modelling code (steps 4–5) is written and unit-tested, waiting only for the real run.

Step 1 — Setup & the leakage firewall

DONE

What we did

- Pulled SA-FARI — gated outlines (Hugging Face) + public frames (Google Cloud).
- Froze the “seen” reference and built the **leakage firewall**: distances only ever see the reference side (with an automated test that a species can’t fake “distance 0” to itself).

Snags & fixes

- **Snag**: the split separates *places* but reuses *species* — not what we assumed. **Fix**: re-planned as **two** experiments (hide a species / hide a place); the frozen model lets us re-slice freely.
- **Snag**: downloading the labels failed with a permissions error (a Hugging Face transfer quirk). **Fix**: switched off the new transfer backend — plain download works.

Step 2 — Measure: frozen SAM 3 → official scorer

DONE

What we did

- Ran frozen SAM 3 over the *whole* test split and scored it with the **official** evaluator, aggregated to one row per (species, place, year).

Result — and why we trust it

pHOTA 0.65 / pDetA 0.52 (detection) / pAssA 0.81 (association). Three independent checks: the numbers are **internally consistent**; they **reproduce the published SAM 3 figures** (our zero-shot + the paper's fine-tuning gain lands on their fine-tuned result); and the model essentially **never invents** an animal that isn't there. **Gate 1 passed.**

Step 3 — The four distances & the feature table DONE / BUILT

What we did

- Built & sanity-checked all four label-free distances (taxonomic, visual, environment, temporal) on real data. DONE
- Built the assembler that joins distances + scores into one modelling table (`features.parquet`); unit-tested. BUILT (awaits the GPU features run.)

Snag & fix — speed

Snag: on Colab, saving thousands of tiny frames to Google Drive was crippling (~ 43 s per clip). **Fix:** downloaded frames **in parallel** to the machine's own fast disk — ~ 4 s per clip, an $\sim 10\times$ speed-up.

Steps 4 & 5 — Fit the law & validate

BUILT

- **Step 4 — fit the law.** The model (predict pDetA/pAssA from the distances, weighted by how much data backs each score) + the “which factor matters?” breakdown are **written and unit-tested on synthetic data.** BUILT
- **Step 5 — out-of-sample check.** Hide whole species / whole places and predict them from distance alone, with honest error bars — **coded and tested.** BUILT

No snags yet — these run on plain CPU and are waiting only for the real `features.parquet`. Running them is **Gate 2**: *does distance really predict a held-out species or place?*

Steps 6 & 7 — Ablations & the reliability tool

NEXT

- **Step 6 — ablations & robustness:** drop each distance to see what it contributes; re-run under alternative choices to confirm the findings hold. NEXT
- **Step 7 — the deployment tool & write-up:** a new (species, place) → a predicted score + a confidence interval; then the dissertation. NEXT

Deliberately deferred until Gate 2 confirms a real signal — no point polishing a tool before we know the law holds.

The biggest practical hurdle — the environment

Snag — Colab kept fighting us

No proper SSH/command-line; it **slept overnight and stopped** mid-run; each fresh session **re-downloaded the 3.4 GB model** and hit the permissions error again; and the GPU limits were too tight for a full run.

Fix — move to a machine we control

A **RunPod GPU over SSH** with a **persistent disk**: no sleep, no re-downloads, resumable long runs, and I can drive the whole pipeline start-to-finish. (Setup captured in a repo runbook.)

Where we stand

The **measurement half is done and validated** (Gate 1) — and it reproduces the published SAM 3 numbers. Every snag along the way is now fixed and captured in code.

Next action: run the distances + fit the law on the SSH machine → Gate 2.

Thank you — questions welcome.